

NCERT Solutions For Class 12 Computer Science

Chapter 8 – Database Concepts

Exercises

1. Give the terms for each of the following:

- a) Collection of logically related records.
- b) DBMS creates a file that contains description about the data stored in the database.
- c) Attribute that can uniquely identify the tuples in a relation.
- d) Special value that is stored when actual data value is unknown for an attribute.
- e) An attribute which can uniquely identify tuples of the table but is not defined as primary key of the table.
- f) Software that is used to create, manipulate and maintain a relational database.

Answer:

- a) Table (Relation)
- b) Data Dictionary (Meta-data)
- c) Primary Key
- d) NULL
- e) Candidate Key / Alternate Key
- f) Database Management System (DBMS)

2. Why foreign keys are allowed to have NULL values? Explain with an example.

Foreign keys are allowed to have NULL values because a record might not always be associated with another table. For example, if a student doesn't have a guardian listed, the GUID (foreign key) in the STUDENT table can be NULL. This means the association is optional for that record.

3. Differentiate between:

- (a) Database state and database schema
- (b) Primary key and foreign key
- (c) Degree and cardinality of a relation

Answer:

S.No	Basis	Term 1	Term 2
(a)	Database State vs Database Schema	Database Schema: It defines the logical structure of the database, including tables, attributes, and relationships. It remains mostly unchanged.	Database State: It represents the actual data in the database at a specific time and keeps changing as data is modified.

(b)	Primary Key vs Foreign Key	Primary Key: A unique identifier for each record in a table. It ensures that no two rows have the same key value.	Foreign Key: A field in one table that refers to the primary key in another table, ensuring referential integrity.
(c)	Degree vs Cardinality of a Relation	Degree: The number of attributes (columns) in a table.	Cardinality: The number of tuples (rows) in a table.

4. Compared to a file system, how does a database management system avoid redundancy in data through a database?

A DBMS stores data centrally in related tables, so repeated information is minimised. Instead of duplicating the same data in different files, linked tables with foreign keys minimise redundancy and ensure changes are reflected everywhere.

5. What are the limitations of file system that can be overcome by a relational DBMS?

Answer:

1. Redundancy and inconsistency
2. Data isolation

3. Difficulty in accessing data

4. Data dependence

5. Poor controlled data sharing

6. A school has a rule that each student must participate in a sports activity. So each one should give only one preference for sports activity. Suppose there are five students in a class, each having a unique roll number. The class representative has prepared a list of sports preferences as shown below. Answer the following:

Table: Sports Preferences

Roll_no	Preference
9	Cricket
13	Football
17	Badminton
17	Football
21	Hockey
24	NULL
NULL	Kabaddi

1. Roll no 24 may not be interested in sports. Can a NULL value be assigned to that student's preference field?

2. Roll no 17 has given two preferences in sports...property violated? Can we use any constraint or key?
3. Kabaddi was not chosen by any student. Is it possible to have this tuple in the Sports Preferences relation?

Answer:

1. In a relational database model where every student must provide exactly one preference for a sports activity, assigning a **NULL** value to Roll No. 24's preference field is **not allowed**, as it violates the rule that each student must have a defined preference.
2. The **primary key constraint** ensures the uniqueness of each record in a relational database table. If Roll No. 17 has two sports preferences, it breaches this rule because a primary key defined on "Roll_no" does not permit duplicate entries. By enforcing a primary key constraint on "Roll_no," the Database Management System (DBMS) automatically prevents such violations by rejecting any attempt to insert multiple rows with the same roll number.
3. Since **no student has chosen Kabaddi** as their preferred sport, a separate entry for Kabaddi should **not exist** in the Sports Preferences table. The table must only contain tuples that represent valid student-sport relationships.

7. In another class having 2 sections, the two respective class representatives have prepared 2 separate Sports Preferences tables, as shown below:

Sports preference of section 1 (arranged on roll number column)

Table: Sports Preferences

Roll_no	Sports
9	Cricket
13	Football
17	Badminton
21	Hockey
24	Cricket

Sports preference of section 2 (arranged on Sports name column, and column order is also different)

Table: Sports Preferences

Sports	Roll_no
Badminton	17
Cricket	9

Cricket	24
Football	13
Hockey	21

Are the states of both the relations equivalent? Justify.

Answer

Yes, both relations represent the same state because the sequence of rows and columns in a relation has no impact on its data. As the information contained in both tables is identical, the two relations are considered equivalent.

8. The school canteen wants to maintain records of items available in the school canteen and generate bills when students purchase any item from the canteen. The school wants to create a canteen database to keep track of items in the canteen and the items purchased by students. Design a database by answering the following questions:

(a) To store each item name along with its price, what relation should be used? Decide appropriate attribute names along with their data type. Each item and its price should be stored only once. What restriction should be used while defining the relation ?

(b) In order to generate bill, we should know the quantity of an item purchased. Should this information be in a new relation or a part of the

previous relation ? If a new relation is required, decide appropriate name and data type for attributes. Also, identify appropriate primary key and foreign key so that the following two restrictions are satisfied:

The same bill cannot be generated for different orders.

Bill can be generated only for available items in the canteen.

(c) The school wants to find out how many calories students intake when they order an item. In which relation should the attribute 'calories' be stored?

Answer

(a) To record each item's name along with its price in the canteen database, a relation (table) named **"Items"** can be created with the following structure:

Items Table

Attributes	Datatype	Constraints
ItemNo	Integer	Primary Key, Unique, Not Null
ItemName	VARCHAR	Not Null
Price	Float	Not Null

The **ItemNo** attribute should be defined as the **primary key** to ensure that each item has a unique identifier. This prevents duplication and

guarantees that every item and its corresponding price appear only once in the database.

(b) The details of items sold should be maintained in a separate relation called "**SaleOrders.**"

SaleOrders Table

Attributes	Datatype	Constraints
OrderNo	Integer	Primary Key, Unique, Not Null
ItemNo	Integer	Foreign Key, Not Null
Quantity	Integer	Not Null
Price	Float	Not Null

This structure satisfies both conditions:

- Each order is uniquely identified by **OrderNo**, ensuring that the same bill cannot be generated multiple times.
- The **ItemNo** attribute acts as a **foreign key**, linking the SaleOrders table to the Items table and ensuring that bills can be generated only for items available in the canteen.

(c) The attribute '**Calories**' should be included in the **Items** table, as it represents a property directly associated with each specific item.

9. An organisation wants to create a database EMPDEPENDENT to maintain following details about its employees and their dependent.

EMPLOYEE(AadharNumber, Name, Address, Department, EmployeeID)

DEPENDENT(EmployeeID, DependentName, Relationship)

(a) Name the attributes of EMPLOYEE, which can be used as candidate keys.

(b) The company wants to retrieve details of dependent of a particular employee. Name the tables and the key which are required to retrieve this detail.

(c) What is the degree of EMPLOYEE and DEPENDENT relation?

Answer:

(a) In the EMPLOYEE table, both AadharNumber and EmployeeID can serve as candidate keys, as each of these attributes can uniquely identify a record within the table.

(b) The EMPLOYEE and DEPENDENT tables are connected through the EmployeeID attribute, which acts as a common key. This key is used to retrieve information about the dependents corresponding to a particular employee.

(c) The EMPLOYEE relation consists of five attributes, giving it a degree of 5, whereas the DEPENDENT relation contains three attributes, resulting in a degree of 3.

10. School uniform is available at M/s Sheetal Private Limited. They have maintained SCHOOL_UNIFORM Database with two relations viz. UNIFORM and COST. The following figure shows database schema and its state.

School Uniform Database

Attributes and Constraints

Table: UNIFORM

Attribute	UCode	UName	UColor
Constraints	Primary Key	Not Null	-

Table: COST

Attribute	UCode	Size	Price
Constraints	Composite Primary Key		>0

Table: UNIFORM

UCode	UName	UColor
1	Shirt	White
2	Pant	Grey
3	Skirt	Grey
4	Tie	Blue
5	Socks	Blue
6	Belt	Blue

Table: COST

UCode	Size	COST Price
1	M	500
1	L	580
1	XL	620
2	M	810
2	L	890
2	XL	940
3	M	770
3	L	830

3	XL	910
4	S	150
4	L	170
5	S	180
5	L	210
6	M	110
6	L	140
6	XL	160

(a) Can they insert the following tuples to the UNIFORM Relation ? Give reasons in support of your answer.

1. 7, Handkerchief, NULL

2. 4, Ribbon, Red

3. 8, NULL, White

(b) Can they insert the following tuples to the COST Relation ? Give reasons in support of your answer.

1. 7, S, 0

2. 9, XL, 100

Answer:

(a)

1. Tuple (7, Handkerchief, NULL): This tuple can be inserted since there is no restriction in the schema that prevents the UColor attribute from having a NULL value.
2. Tuple (4, Ribbon, Red): This tuple is valid for insertion as all attributes contain appropriate non-null values that comply with the schema constraints.
3. Tuple (8, NULL, White): This tuple cannot be inserted because the UName attribute is defined as NOT NULL in the schema, and therefore, it must always contain a value.

(b)

1. Tuple (7, S, 0): This tuple is not allowed to be inserted because the Price attribute must have a value greater than 0, as specified in the schema constraints.
2. Tuple (9, XL, 100): This tuple can be successfully inserted since all attributes have valid, non-null values, and the Price satisfies the condition of being greater than 0.

11. In a multiplex, movies are screened in different auditoriums. One movie can be shown in more than one auditorium. In order to maintain the record of movies, the multiplex maintains a relational database consisting of two relations viz. MOVIE and AUDI respectively as shown below :

Movie(Movie_ID, MovieName, ReleaseDate)

Audi(AudiNo, Movie_ID, Seats, ScreenType, TicketPrice)

(a) Is it correct to assign Movie_ID as the primary key in the MOVIE relation ? If no, then suggest an appropriate primary key.

(b) Is it correct to assign AudiNo as the primary key in the AUDI relation ? If no, then suggest appropriate primary key.

(c) Is there any foreign key in any of these relations ?

Answer:

Answer

(a) Yes, assigning **Movie_ID** as the **primary key** in the **MOVIE** relation is appropriate since every movie possesses a unique **Movie_ID** that distinguishes it from others.

(b) It is not suitable to designate **AudiNo** as the primary key in the **AUDI** relation because the same auditorium number can be associated with different movies shown at various times. To ensure that each record is uniquely identified, a **composite primary key** made up of **AudiNo** and **Movie_ID** should be defined.

(c) Yes, the **AUDI** relation includes a **foreign key**. The **Movie_ID** attribute in the **AUDI** table acts as a foreign key that references the **Movie_ID** primary key in the **MOVIE** relation, thereby establishing a link between the two tables.

12. For the below given database STUDENT-PROJECT, answer the following:

(a) Name primary key of each table.

(b) Find foreign key(s) in table PROJECT-ASSIGNED.

(c) Is there any alternate key in table STUDENT? Give justification for your answer.

(d) Can a user assign duplicate value to the field RollNo of STUDENT table? Justify.

Student Project Database

Table: STUDENT

Roll No	Name	Class	Section	Registration_ID
11	Mohan	XI	1	IP-101-15
12	Sohan	XI	2	IP-104-15
21	John	XII	1	CS-103-14
22	Meena	XII	2	CS-101-14
23	Juhi	XII	2	CS-101-10

Table: PROJECT

ProjectNo	PName	SubmissionDate
101	Airline Database	12/01/2018
102	Library Database	12/01/2018
103	Employee Database	15/01/2018
104	Student Database	12/01/2018
105	Inventory Database	15/01/2018
106	Railway Database	15/01/2018

Table: PROJECT ASSIGNED

Registration_ID	ProjectNo
IP-101-15	101
IP-104-15	103
CS-103-14	102
CS-101-14	105
CS-101-10	104

Answer:

(a) The primary key for each table is as follows:

- **STUDENT:** Roll No
- **PROJECT:** ProjectNo
- **PROJECT_ASSIGNED:** Registration_ID

(b) The ProjectNo column in the PROJECT_ASSIGNED table functions as a foreign key, linking it to the ProjectNo column in the PROJECT table to

establish a relationship between the two.

(c) In the STUDENT table, the Registration_ID attribute acts as an alternate key because it uniquely identifies every student in addition to the primary key.

(d) Duplicate values cannot be assigned to the Roll No field in the STUDENT table, as it serves as the primary key, which enforces uniqueness across all records.

Answer:

(a) No, a record cannot be inserted with a missing Roll No value in the STUDENT table because the Roll No attribute is defined as both Primary Key and NOT NULL. Inserting a null value would violate these constraints, making the operation invalid.

(b) Yes, a record can be inserted without a Registration_ID in the STUDENT table, as this attribute does not have a NOT NULL constraint defined in the schema. Therefore, it is permissible to leave this field blank.

(c) Yes, inserting a project record without a SubmissionDate is allowed because the attribute does not have a NOT NULL constraint in the PROJECT table. Hence, a project detail can exist even if its submission date is not provided.

(d) No, this operation cannot be executed because ProjectNo in the PROJECT_ASSIGNED table acts as a foreign key referencing the Project table's primary key. Since the value 206 does not exist in the PROJECT table, inserting it would violate referential integrity and is therefore not allowed.